## AD 2. AERODROMES

### VEIM AD 2.1 AERODROME LOCATION INDICATOR AND NAME

VEIM - IMPHAL AIRPORT, IMPHAL

### VEIM AD 2.2 AERODROME GEOGRAPHICAL AND ADMINISTRATIVE DATA

| | | | |
|---|---|---|---|
| 1 | Aerodrome reference point coordinates and its site | 244552N 0935359E<br>M. Brg. 221.87 DEG/813M from physical extremity of RWY 22 | |
| 2 | Direction and distance of aerodrome reference point from the center of the city or town which the aerodrome serves | 225 DEG/7KM from Imphal city | |
| 3 | Aerodrome elevation and reference temperature | 2544 FT / 32.0 DEG C | |
| 4 | Magnetic variation, date of information and annual change | 0.58 DEG W (2010) /0.017 DEG W | |
| 5 | Name of aerodrome operator, address, telephone, telefax, e-mail address, AFS address, website (if available) | The Airport Director<br>Airports Authority of India,<br>Imphal Airport,<br>Tulihal, Imphal - 795140 | |
| | | Telephone: | (91) 3852455138<br>(91) 3852455153<br>(91) 9436237721 |
| | | Fax: | (91) 3852455076<br>(91) 3852455153<br>(91) 3852455138 |
| | | AFS: | VEIMYDYX |
| | | Email: | apdimphal@aai.aero |
| 6 | Types of traffic permitted (IFR/VFR) | IFR/VFR | |
| 7 | Remarks | Elevation in EGM08 | |

### VEIM AD 2.3 OPERATIONAL HOURS

| | | | |
|---|---|---|---|
| 1 | Aerodrome Operator | MON-FRI 0400-1230 UTC (0930-1800 IST)<br>SAT, SUN+HOL: NIL | |
| 2 | Custom and immigration | HO | |
| 3 | Health and sanitation | One doctor and a staff nurse is provided by Manipur Health Department, Govt of Manipur at terminal building during operation hours | |
| 4 | AIS briefing office | As ATS | |
| 5 | ATS reporting office (ARO) | As ATS | |
| 6 | MET Briefing office | As ATS | |
| 7 | Air Traffic Service | Consult NOTAM For Current ATS HR. | |
| 8 | Fuelling | As ATS | |
| 9 | Handling | As ATS | |
| 10 | Security | As ATS | |
| 11 | De-icing | NIL | |
| 12 | Remarks | Outside of ATS hour's services are available O/R with 24 HR PN to AD. | |
| | | ATS approved hourly runway traffic handling capacity | Maximum number of arrival and departure- 08<br>(minimum spacing between two successive arrivals shall be more than five minutes)<br>Maximum number of arrival only – 06<br>Maximum number of departure only -10 |

## VEIM AD 2.4 HANDLING SERVICES AND FACILITIES

| | | |
|---|---|---|
| 1 | Cargo-handling facilities | With Prior Arrangement with Local Airlines |
| 2 | Fuel and Oil types | JET A1 |
| 3 | Fuelling facilities and capacity | 4 Bowsers Capacity 11KL,11KL,16 KL, 06 KL (IOCL) and 2 bowsers for BPCL of 15 KL |
| 4 | De-icing facilities | NIL |
| 5 | Hangar space for visiting aircraft | NIL |
| 6 | Repair facilities for visiting aircraft | NIL |
| 7 | Remarks | Refuelling facility AVBL at Imphal for Airlines operator. |

## VEIM AD 2.5 PASSENGER FACILITIES

| | | |
|---|---|---|
| 1 | Hotel(s) at or in the vicinity of aerodrome | In the City |
| 2 | Restaurant(s) at or in the vicinity of aerodrome | In city only |
| 3 | Transportation possibilities | Taxi /Car on Hire from AD. |
| 4 | Medical Facilities | First aid At AD. Hospitals in City. |
| 5 | Bank and post office at or in the vicinity of aerodrome | Banks: In the city<br>Post office: 0930-1600 IST, except Sun & Holiday. |
| 6 | Tourist office | AVBL 1000-1500 IST, except Sun & Holiday |
| 7 | Remarks | NIL |

## VEIM AD 2.6 RESCUE AND FIRE FIGHTING SERVICES

| | | |
|---|---|---|
| 1 | Aerodrome category for fire fighting | Within ATS HR: CAT-7 |
| 2 | Rescue equipment | Available as per category. |
| 3 | Capability for removal of disabled aircraft | 1.The critical aircraft for the airport is A-321.<br>2. Primary responsibility for removal of disabled aircraft rests with the concerned airline.<br>3. The airlines may contact ‘Aerodrome Coordinator for Disabled Aircraft Removal Operations’ for coordinating with the agencies having Disabled Aircraft Recovery Kit.<br>4. Contact Details of Aerodrome Coordinator for Disabled Aircraft Removal Operations:<br>Airport Director<br>Contact number :0385-2455069<br>Alternate Coordinator: 0385-2455138<br>E-mail:- apdimphal@aai.aero |
| 4 | Remarks | |

## VEIM AD 2.7 SEASONAL AVAILABILITY CLEARING

| | | |
|---|---|---|
| 1 | Type(s) of clearing equipment | NIL |
| 2 | Clearance priorities | NIL |
| 3 | Remarks | NIL |

## VEIM AD 2.8 APRONS, TAXIWAYS AND CHECK LOCATIONS/POSITIONS DATA

| | | |
|---|---|---|
| 1 | Designation, surface and strength of aprons | Refer Aircraft Parking/Docking Chart. |
| 2 | Designation, width, surface and strength of taxiways | Refer Aircraft Parking/Docking Chart. |
| 3 | Location and elevation of altimeter checkpoints | At Apron<br>Refer Aerodrome Chart for location. |

| | | |
|---|---|---|
| 4 | Location of VOR checkpoints | TWY B: 244548.33N 0935355.92E<br>Elevation: 2540FT<br>TWY C: 244544.35N 0935352.91E<br>Elevation: 2538FT |
| 5 | Position of INS checkpoints | NIL |
| 6 | Remarks | i. Coordinates of RWY Holding Positions:<br>a) TWY A: 244601.70N 0935406.40E, Elev. 2376 FT<br>b) TWY B : 244548.15N 0935356.21E, Elev. 2376 FT<br>c) TWY C: 244544.17N 0935353.19E, Elev. 2375 FT<br>d) TWY C1: 244540.93N 0935350.75E, Elev. 2373 FT<br>e) TWY D: 244536.24N 0935347.209E, Elev. 275 FT<br>ii. Coordinates of RWY & Taxiway Intersections:<br>a) TWY A: 244559.98N 09354 09.14E, Elev. 2376 FT<br>b) TWY B: 244546.47N 0935358.91E, Elev. 2376 FT<br>c) TWY C: 244542.46N 0935355.90E, Elev. 2376 FT<br>d) TWY C1: 244539.22N 0935353.45E, Elev. 2376 FT<br>e) TWY D: 244534.54N 0935349.91E, Elev. 2376 FT<br>iii. Parking Limitation & Restriction:<br>a) The aircraft parked at Stand No. 1 cannot taxi out when Stand No. 2 is occupied.<br>b) When Stand No. 1 is occupied, ACFT cannot Taxi-in to Stand No. 2.<br>c) Five aircraft can be parked simultaneously due configuration of aircraft stands.<br>d) Apron lights for stands 1, 2, 3, 4, 5, & 6 available.<br>e) Prior approval to be taken before releasing any aircraft due shortage of aircraft stands.<br>iv. Elevations in EGM08 |

## VEIM AD 2.9 SURFACE MOVEMENT GUIDANCE AND CONTROL SYSTEM AND MARKINGS

| | | |
|---|---|---|
| 1 | Use of aircraft stand identification signs, taxiway guidelines and visual docking/parking guidance system at aircraft stands | Taxiing guidance provided on RTF<br>Guidelines at Apron. |
| 2 | Runway and taxiway markings and lights | RWY Markings: Designation, THR, TDZ, Centreline, Edge, RWY End.<br>RWY Lights: Edge, RWY End, THR<br>TWY Markings: Centreline, Holding Positions<br>TWY Lights: Edge |
| 3 | Stop bars (if any) | NIL |
| 4 | Remarks | NIL |

## VEIM AD 2.10 AERODROME OBSTACLES

| In Approach/Take-off/Circling Area and at AD | | | | | |
|---|---|---|---|---|---|
| **1** | **2** | **3** | **4** | **5** | **6** |
| RWY/Area affected | Obstacle type | Coordinates | Elevation (EGM-08) | Marking/LGT | Remarks |
| 22/TKOF 04/APCH | BUILDING | 244100.7N 0935014.7E | 3093 FT | NO | SECURITY HUT |
| 22/TKOF 04/APCH | OTHER | 244058.9N 0935016.5E | 3118 FT | NO | SOLAR OBSTRUCTION LIGHT ON HILL |
| 22/TKOF 04/APCH | BUILDING | 244452.9N 0935325.0E | 2557 FT | NO | SECURITY HUT |
| 22/APCH 04/TKOF | BUILDING | 244615.0N 0935427.1E | 2563 FT | NO | SECURITY HUT |
| 22/APCH 04/TKOF | POLE | 244624.6N 0935420.0E | 2574 FT | NO | LIGHT POLE |
| 22/APCH 04/TKOF | POLE | 244626.4N 0935422.0E | 2574 FT | NO | LIGHT POLE |
| 22/APCH 04/TKOF | POLE | 244626.6N 0935423.0E | 2572 FT | NO | ELECTRIC POLE |
| 22/APCH 04/TKOF | BUILDING | 244626.3N 0935423.6E | 2571 FT | NO | BUILDING |
| 22/APCH 04/TKOF | ANTENNA | 244627.3N 0935424.1E | 2605 FT | NO | CELLPHONE MAST ON BUILDING |
| 22/APCH 04/TKOF | BUILDING | 244627.4N 0935424.2E | 2576 FT | NO | BUILDING |
| 22/APCH 04/TKOF | BUILDING | 244627.9N 0935425.8E | 2576 FT | NO | BUILDING |
| 22/APCH 04/TKOF | BUILDING | 244630.0N 0935423.1E | 2590 FT | NO | BUILDING |
| 22/APCH 04/TKOF | BUILDING | 244629.0N 0935425.9E | 2588 FT | NO | BUILDING UNDER CONSTRUCTION |
| 22/APCH 04/TKOF | BUILDING | 244630.4N 0935424.4E | 2606 FT | NO | BUILDING |
| 22/APCH 04/TKOF | ANTENNA | 244630.5N 0935424.6E | 2611 FT | NO | CELLPHONE MAST ON BUILDING |
| 22/APCH 04/TKOF | BUILDING | 244629.2N 0935428.7E | 2592 FT | NO | BUILDING |
| 22/APCH 04/TKOF | BUILDING | 244646.4N 0935439.2E | 2622 FT | NO | TEMPLE |
| 22/APCH 04/TKOF | ANTENNA | 244653.7N 0935447.0E | 2682 FT | NO | CELLPHONE MAST |
| 22/APCH 04/TKOF | ANTENNA | 244657.8N 0935445.7E | 2688 FT | NO | CELLPHONE MAST |
| 22/APCH 04/TKOF | ANTENNA | 244706.4N 0935500.1E | 2715 FT | NO | CELLPHONE MAST |
| 22/APCH 04/TKOF | ANTENNA | 244707.4N 0935502.2E | 2684 FT | NO | CELLPHONE MAST |
| In circling area and at AD | BUILDING | 244511.6N 0935325.7E | 2559 FT | NO | SECURITY HUT |
| In circling area and at AD | BUILDING | 244504.3N 0935334.0E | 2562 FT | NO | SECURITY HUT |
| In circling area and at AD | BUILDING | 244504.3N 0935333.9E | 2562 FT | NO | SECURITY HUT |

| In Approach/Take-off/Circling Area and at AD | | | | | |
|---|---|---|---|---|---|
| **1** | **2** | **3** | **4** | **5** | **6** |
| RWY/Area affected | Obstacle type | Coordinates | Elevation (EGM-08) | Marking/LGT | Remarks |
| In circling area and at AD | NAVAID | 244503.5N 0935331.7E | 2563 FT | NO | GLIDE PATH DME |
| In circling area and at AD | BUILDING | 244503.5N 0935331.7E | 2552 FT | NO | GLIDE PATH HUT |
| In circling area and at AD | NAVAID | 244503.4N 0935331.6E | 2587 FT | NO | GLIDE PATH MAIN ANTENNA |
| In circling area and at AD | NAVAID | 244501.8N 0935330.6E | 2554 FT | NO | GLIDE PATH MONITOR ANTENNA |
| In circling area and at AD | BUILDING | 244501.3N 0935317.5E | 2561 FT | NO | SECURITY HUT |
| In circling area and at AD | BUILDING | 244500.9N 0935318.0E | 2549 FT | NO | SECURITY HUT |
| In circling area and at AD | ANTENNA | 244424.9N 0935224.5E | 2708 FT | NO | CELLPHONE MAST |
| In circling area and at AD | ANTENNA | 244603.0N 0935418.5E | 2567 FT | NO | SECURITY HUT |
| In circling area and at AD | TREE | 244618.3N 0935414.2E | 2577 FT | NO | TREE |
| In circling area and at AD | TREE | 244619.9N 0935412.8E | 2603 FT | NO | GROUP OF TREES |
| In circling area and at AD | POLE | 244623.4N 0935418.6E | 2574 FT | NO | LIGHT POLE |
| In circling area and at AD | ANTENNA | 244642.8N 0935515.2E | 2715 FT | NO | CELLPHONE MAST |
| In circling area and at AD | ANTENNA | 244547.8N 0935328.1E | 2717 FT | NO | CELLPHONE MAST |
| In circling area and at AD | TREE | 244527.6N 0935337.8E | 2569 FT | NO | GROUP OF TREES |
| In circling area and at AD | BUILDING | 244517.9N 0935344.0E | 2564 FT | NO | SECURITY HUT |
| In circling area and at AD | TANK | 244550.2N 0935346.9E | 2651 FT | NO | OVER HEAD WATER TANK |
| In circling area and at AD | OTHER | 244548.1N 0935348.5E | 2613 FT | NO | BEACON |
| In circling area and at AD | OTHER | 244548.2N 0935348.5E | 2624 FT | NO | LIGHTNING ARRESTOR |
| In circling area and at AD | OTHER | 244551.8N 0935350.9E | 2615 FT | NO | LIGHTNING ARRESTOR |
| In circling area and at AD | NAVAID | 244527.3N 0935351.1E | 2568 FT | NO | DVOR MONITOR ANTENNA |
| In circling area and at AD | BUILDING | 244551.8N 0935351.4E | 2611 FT | NO | TERMINAL BUILDING |
| In circling area and at AD | NAVAID | 244529.6N 0935351.5E | 2569 FT | NO | DVOR DME |
| In circling area and at AD | BUILDING | 244530.0N 0935351.7E | 2548 FT | NO | DVOR BUILDING |
| In circling area and at AD | NAVAID | 244529.9N 0935351.8E | 2562 FT | NO | DVOR MAIN ANTENNA |
| In circling area and at AD | OTHER | 244530.1N 0935352.1E | 2575 FT | NO | LIGHTNING ARRESTOR |
| In circling area and at AD | BUILDING | 244530.9N 0935354.2E | 2561 FT | NO | SECURITY HUT |

| In Approach/Take-off/Circling Area and at AD | | | | | |
|---|---|---|---|---|---|
| **1** | **2** | **3** | **4** | **5** | **6** |
| RWY/Area affected | Obstacle type | Coordinates | Elevation (EGM-08) | Marking/LGT | Remarks |
| In circling area and at AD | BUILDING | 244530.9N 0935354.2E | 2561 FT | NO | SECURITY HUT |
| In circling area and at AD | BUILDING | 244555.3N 0935357.6E | 2569 FT | NO | IOCL SECURITY HUT |
| In circling area and at AD | OTHER | 244556.3N 0935358.9E | 2569 FT | NO | W.D.I. |
| In circling area and at AD | POLE | 244556.5N 0935359.1E | 2574 FT | NO | LIGHT POLE |
| In circling area and at AD | POLE | 244554.3N 0935359.5E | 2575 FT | NO | LIGHT POLE |
| In circling area and at AD | NAVAID | 244554.5N 0935359.7E | 2546 FT | NO | MET INSTRUMENT |
| In circling area and at AD | POLE | 244557.8N 0935400.2E | 2568 FT | NO | LIGHT POLE |
| In circling area and at AD | POLE | 244556.1N 0935400.3E | 2569 FT | NO | LIGHT POLE |
| In circling area and at AD | OTHER | 244540.6N 0935400.7E | 2566 FT | NO | W.D.I. |
| In circling area and at AD | POLE | 244557.2N 0935401.2E | 2568 FT | NO | LIGHT POLE |
| In circling area and at AD | NATURAL HIGHPOINT | 244558.0N 0935402.4E | 2549 FT | NO | GROUND LEVEL |
| In circling area and at AD | BUILDING | 244541.7N 0935402.4E | 2562 FT | NO | SECURITY HUT |
| In circling area and at AD | OTHER | 244559.9N 0935403.9E | 2553 FT | NO | FENCING POLE |
| In circling area and at AD | BUILDING | 244551.9N 0935409.5E | 2561 FT | NO | SECURITY HUT |
| In circling area and at AD | NATURAL HIGHPOINT | 244740.5N 0934504.4E | 5020 FT | NO | HILL |
| In circling area and at AD | NATURAL HIGHPOINT | 245002.4N 0935340.5E | 3661 FT | NO | HILL |
| In circling area and at AD | NATURAL HIGHPOINT | 245053.5N 0935345.6E | 3648 FT | NO | HILL |
| In circling area and at AD | NATURAL HIGHPOINT | 245049.6N 0935431.5E | 3570 FT | NO | HILL |
| In circling area and at AD | NATURAL HIGHPOINT | 245025.3N 0935558.6E | 3051 FT | NO | HILL |
| In circling area and at AD | NATURAL HIGHPOINT | 245405.3N 0935619.7E | 3471 FT | NO | HILL |
| In circling area and at AD | NATURAL HIGHPOINT | 245249.6N 0935715.6E | 3281 FT | NO | HILL |
| In circling area and at AD | NATURAL HIGHPOINT | 244117.8N 0935806.0E | 3402 FT | NO | HILL |
| In circling area and at AD | NATURAL HIGHPOINT | 244155.8N 0935814.6E | 3671 FT | NO | HILL |
| In circling area and at AD | NATURAL HIGHPOINT | 244309.9N 0935843.8E | 3346 FT | NO | HILL |
| In circling area and at AD | NATURAL HIGHPOINT | 244549.3N 0935944.6E | 3507 FT | NO | HILL |
| In circling area and at AD | NATURAL HIGHPOINT | 245038.6N 0940036.0E | 3287 FT | NO | HILL |

| In Approach/Take-off/Circling Area and at AD | | | | | |
|---|---|---|---|---|---|
| **1** | **2** | **3** | **4** | **5** | **6** |
| RWY/Area affected | Obstacle type | Coordinates | Elevation (EGM-08) | Marking/LGT | Remarks |
| In circling area and at AD | NATURAL HIGHPOINT | 244024.1N 0940031.3E | 3415 FT | NO | HILL |
| In circling area and at AD | NATURAL HIGHPOINT | 244831.7N 0940037.1E | 3209 FT | NO | HILL |
| In circling area and at AD | NATURAL HIGHPOINT | 244118.9N 0940037.0E | 3153 FT | NO | HILL |
| In circling area and at AD | NATURAL HIGHPOINT | 244238.7N 0940038.3E | 3209 FT | NO | HILL |
| In circling area and at AD | NATURAL HIGHPOINT | 244441.0N 0940045.8E | 3579 FT | NO | HILL |
| In circling area and at AD | NATURAL HIGHPOINT | 245116.3N 0940110.8E | 3071 FT | NO | HILL |
| In circling area and at AD | NATURAL HIGHPOINT | 244649.0N 0940114.7E | 5266 FT | NO | HILL TOP |
| In circling area and at AD | NATURAL HIGHPOINT | 244650.5N 0940206.7E | 5039 FT | NO | HILL |

## VEIM AD 2.11 METEOROLOGICAL INFORMATION PROVIDED

| | | |
|---|---|---|
| 1 | Name of the associated meteorological office | Imphal |
| 2 | Hours of service and, where applicable, the designation of the responsible meteorological office outside these hours | As ATS |
| 3 | Office responsible for preparation of TAFs and periods of validity and interval of issuance of the forecasts | Guwahati<br>9 HR [00-09] |
| 4 | Availability of the trend forecast for the aerodrome and interval of issuance | Available at 30 minutes interval |
| 5 | Information on how briefing and/or consultation is provided | NIL |
| 6 | Types of flight documentation supplied and language(s) used in flight documentation | Tabular Form English |
| 7 | Charts and other information displayed or available for briefing or consultation | NIL |
| 8 | Supplementary equipment available for providing information on meteorological conditions, e.g. weather radar and receiver for satellite images; | Current Weather Instrument System (CWIS) Available With The Following Components:<br>1. Anemometer And Wind Vane.<br>2. Thermistor/ Hygristor and Baro Switch |
| 9 | The air traffic services unit(s) provided with meteorological information | Imphal ATC and ACS |
| 10 | Additional information, e.g. concerning any limitation of service. | Documents available on request.<br>Transmissometer for RVR AVBL at RWY04 TDZ |

## VEIM AD 2.12 RUNWAY PHYSICAL CHARACTERISTICS

| Designations | TRUE Bearings | Dimensions of RWY (M) | Strength of pavement (PCN) and associated data) and surface of runway and associated stopways | Geographical coordinates for threshold and runway end |
|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 |
| 04 | 34.61 DEG | 2744 x 45 M | 67/F/D/W/T Macadam | THR: 244458.02N 0935322.31E RWY END: 244611.42N 0935417.79E |
| 22 | 214.61 DEG | 2744 x 45 M | 67/F/D/W/T Macadam | THR: 244611.42N 0935417.79E RWY END: 244458.02N 0935322.31E |

| THR elevation and highest elevation of TDZ of precision APP RWY | Slope of runway and associated stopway | Dimensions of stopway (M) | Dimensions of clearway (M) | Dimensions of strips (M) |
|---|---|---|---|---|
| 6 | 7 | 8 | 9 | 10 |
| THR: 2538.0FT TDZ: 2543.0FT | 0.06% | NIL | | 2864 x 150 M |
| THR: 2544.0FT TDZ: 2544.0FT | -0.06% | NIL | | 2864 x 150 M |

| Dimensions of runway end safety areas | Location and description of arresting system (if any) | Existence of an obstacle-free zone | Remarks. |
|---|---|---|---|
| 11 | 12 | 13 | 14 |
| 240M x 90M | NIL | | 1. Aircraft to make 180 Degree turn at turn pad by using turn pad markings to avoid damage to RWY Surface. 2. Elevation in EGM 08. |
| 240M x 90M | NIL | | 1. Aircraft to make 180 Degree turn at turn pad by using turn pad markings to avoid damage to RWY Surface. 2. Elevation in EGM 08 |

## VEIM AD 2.13 DECLARED DISTANCES

| RWY Designator | Take-off run available TORA (M) | Take-off distance available TODA (M) | Accelerate distance available ASDA (M) | Landing distance available LDA (M) | Remarks (including runway entry or start point where alternative reduced declared distances have been declared) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| 04 | 2744 | 2744 | 2744 | 2744 | |
| 22 | 2744 | 2744 | 2744 | 2744 | |

## VEIM AD 2.14 APPROACH AND RUNWAY LIGHTING

| Runway Designator | Type, length and intensity of approach lighting system | Runway threshold lights, colour and wing bars | Type of visual slope indicator system | Length of runway touchdown zone lights |
|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 |
| **04** | CAT I<br>900 M<br>LIH | Green | PAPI<br>LEFT/3.50 DEG<br>MEHT<br>(59.71FT) | NIL |
| **22** | SALS<br>420 M<br>LIH | Green | PAPI<br>LEFT/3.50 DEG<br>MEHT<br>(58.11FT) | NIL |

| Length, spacing, colour and intensity of runway centre line lights | Length, spacing, colour and intensity of runway edge lights | Colour of runway end lights and wing bars | Length and colour of stopway lights | Remarks |
|---|---|---|---|---|
| 6 | 7 | 8 | 9 | 10 |
| | 2760 M<br>60 M<br>White<br>LIH | Red | NIL | NIL |
| | 2760 M<br>60 M<br>White<br>LIH | Red | NIL | NIL |

## VEIM AD 2.15 OTHER LIGHTING, SECONDARY POWER SUPPLY

| | | | |
|---|---|---|---|
| 1 | Location, characteristics and hours of operation of aerodrome beacon/identification beacon (if any) | ABN | At Tower Building, FLG W&G EV2SEC, As ATS HR |
| | | IBN | NIL |
| 2 | Location and lighting (if any) of anemometer/ landing direction indicator; | LDI | Lighted, NE corner of Apron |
| | | Anemometer | On TWR Bldg, Lighted |
| 3 | Taxiway edge and taxiway centre line lights; | Edge | All TWY except TWY 'A' |
| | | Centre Line | --- |
| 4 | Secondary power supply including switch-over time; | 1) Secondary Power Supply to RWY EDGE lights,RWY Turn Pad lights,PAPI lights,Approach lights,TWY lights,Apron lights.Switch over time 1 sec.<br>2) Secondary Power Supply to all other lights switch over time 15 Sec. | | |
| 5 | Remarks | WDI Lighted<br>Road HLDG LGT AVBL at access rod of RWY04, RWY22, Fire Station, Transmitter road. | | |

## VEIM AD 2.16 HELICOPTER LANDING AREA

| | | |
|---|---|---|
| 1 | Geographical coordinates of the geometric centre of touchdown and lift-off (TLOF) or of each threshold of final approach and take-off (FATO) area | Not Established |
| 2 | TLOF and/or FATO area elevation: | Not Established |
| 3 | TLOF and FATO area dimensions to the nearest metre or foot, surface type, bearing strength and marking; | Not Established |
| 4 | True bearings of FATO; | Not Established |
| 5 | Declared distances available | Not Established |
| 6 | Approach and FATO lighting; | Not Established |
| 7 | Remarks | Not Established |

## VEIM AD 2.17 AIR TRAFFIC SERVICE AIRSPACE

| | | |
|---|---|---|
| 1 | Airspace designation, geographical coordinates and lateral limits | CTR: Area bounded by lines joining points 250002N 0932548E; 245902N 0933048E then along the clockwise arc of a circle of 25NM radius centred on 244530N 0935352E to 243902N 0932748E; 244102N 0932148E then along the counter clockwise arc of a circle of 25NM radius centred on 245449N 0925840E to point of origin. |
| 2 | Vertical limits | FL 200 |
| 3 | Airspace classification | D |
| 4 | Call sign and language(s) of the air traffic services unit providing service; | Imphal Tower / Imphal Approach, English |
| 5 | Transition altitude | 11000 FT |
| 6 | Hours of applicability | HO |
| 7 | Remarks | NIL |

## VEIM AD 2.18 AIR TRAFFIC SERVICES COMMUNICATION FACILITIES

| Service Designation | Call sign | Channel(s) | SATVOICE Number(s), if available | Logon address, as appropriate | Hours of operation | Remarks |
|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| APP | Imphal Approach | 118.550 MHZ | | | HO | Backup FREQ |
| APP | Imphal Approach | 124.350 MHZ | | | HO | NIL |
| TWR | Imphal Tower | 118.550 MHZ | | | HO | BACKUP FREQ. |
| TWR | Imphal Tower | 124.350 MHZ | | | | SMC AND APP SERVICES COMBINED WITH TWR. |
| ATIS | Imphal Imnformation | 126.650 MHZ | | | | NIL |
| ALRS | EMERGENCY FREQUENCY | 121.500 MHZ | | | | |

| Logon address, as appropriate | Hours of operation | Remarks |
|---|---|---|
| | HO | NIL |

## VEIM AD 2.19 RADIO NAVIGATION AND LANDING AIDS

| Type of aids, magnetic variation and type of supported operation for ILS/ MLS, basic GNSS, SBAS and GBAS, and for VOR/ILS/MLS station used for technical lineup of the aid | Identification | Frequency(ies), Channel number(s), Service provider, and reference path identifier(s) (RPI), as appropriate | Hours of operation, as appropriate; |
|---|---|---|---|
| 1 | 2 | 3 | 4 |
| LOC 04<br>CAT I | IIPH | 110.300 MHz | As ATS |
| GP 04 | | 335.000 MHz | As ATS |
| DME ILS 04 | IIPH | CH40X | As ATS |
| DVOR/DME | IIM | 115.900 MHz<br>CH106X | As ATS |

| Geographical coordinates of the position of the transmitting antenna | Elevation of transmitting antenna of DME/ elevation of GBAS reference point | Service volume radius from the GBAS reference point | Remarks |
|---|---|---|---|
| 5 | 6 | 7 | 8 |
| 244621.7N<br>0935425.8E | | | ILS CAT I |
| 244503.3N<br>0935331.7E | | | 3.5 DEG |
| 244503.3N<br>0935331.7E | 2563 FT | | 1. Collocated with GP 04<br>2. Elevation in EGM08 |
| 244447.3N<br>0935325.3E | 2568 FT | | 1. Co-located with DVOR<br>2. Vertical datum: EGM08 |

## VEIM AD 2.20 LOCAL AERODROME REGULATIONS

• No Aircraft shall take lock turn/ pivotal turn on the Runway to avoid pavement damage.

## LOW VISIBILITY PROCEDURES, IMPHAL AIRPORT, IMPHAL (VEIM)

1.INTRODUCTION

1.1Until the latest amendment of DGCA Civil Aviation Requirements (CAR) Section 8, Series 'C', Part-I on All-Weather Operations, Low visibility procedures were required at aerodromes for the purpose of ensuring safe operations during Categories II and III approaches and/or low visibility take-offs (LVTO).

1.2However, in latest amendment to CAR (Rev. 10), para 5.3, following provision regarding Low visibility procedures is added. "An operator shall not conduct takeoff with RVR/visibility less than standard Category I conditions of 550m RVR/800 m visibility unless low visibility procedures are enforced".

1.3This provision necessitated the need for low visibility procedures for accommodating/permitting departures in Visibility/ RVR less than 800M/550M even at such airports where there are no CAT II and CAT III operations.

1.4As per provisions of Aerodrome Design and Operations CAR and ICAO Annex 14, Runway centre line lights are required for take-off in RVR below 400 M. Further, as per Aerodrome Design and operations CAR para 9.8.7, Surface Movement Radar (SMR) need to be provided at an aerodrome intended for use in Runway Visual Range conditions less than a value of 350 M

1.5Accordingly, Low visibility Procedures have been developed for Imphal Airport to accommodate/permit departures in Visibility/RVR less than 800M/550M from RWY 04 (runway served with RVR instruments). and since, there is No SMR facility at Imphal Airport, these procedures will be applicable for take-off in VIS/RVR below 800M/550M but not less than 400 M RVR.

2PURPOSE

2.1The purpose of this document is to define the Standard Operating Procedures for Low Visibility operations at Imphal Airport.

3SCOPE

3.1 This is applicable to all the concerned personnel involved in the Low Visibility Operation as mentioned in this SOP.

3.2 The Low Visibility Procedure (LVP) describe the procedures and actions that are required to be taken by the Air Traffic Control (ATC), Pilots, Airline Operators, MET dept, Vehicle Operators, Airport Operations, Electrical Engg., CNS, Airport Fire Services and Ground Handling Agencies during Low Visibility Take-Offs at Imphal Airport

3.3 The procedures contained in this document shall be read in conjunction with other applicable ICAO DOCs and Annexes / DGCA CARs and relevant circulars, AAI CHQ/RHQ instructions on the subject issued from time to time.

4 DEFINITIONS

4.1 Aerodrome Operating Minima: The limits of usability of an aerodrome for:

a) take off, expressed in terms of runway visual range and/or visibility and, if necessary, cloud conditions.

b) landing in 2D instrument approach operations, expressed in terms of visibility and/or runway visual range; minimum descent altitude/height (MDA/H) and, if necessary, cloud conditions; and

c) landing in 3D instrument approach operations, expressed in terms of visibility and/or runway visual range and decision altitude/height (DA/H) appropriate to the type and/or category of the operation

4.2 Instrument approach operations (Annex 6): An approach and landing using instruments for navigation guidance based on an instrument approach procedure. There are two methods for executing instrument approach operations:

a) a two-dimensional (2D) instrument approach operation, using lateral navigation guidance only; and

b) a three-dimensional (3D) instrument approach operation, using both lateral and vertical navigation guidance

Note – Lateral and vertical navigation guidance refers to the guidance provided either by:

a) a ground-based radio navigation aid; or

b) computer-generated navigation data from ground-based, space-based, self-contained navigation aids or a combination of these

Instrument approach operations shall be classified based on the designed lowest operating minima below which an approach operation shall only be continued with the required visual reference as follows:

a) Type A: a minimum descent height or decision height at or above 75 m (250 ft); and

b) Type B: a decision height below 75 m (250 ft). Type B instrument approach operations are categorized as:

i) Category I (CAT I): a decision height not lower than 60 m (200 ft) and with either a visibility not less than 800 m or a runway visual range not less than 550 m;

ii) Category II (CAT II): a decision height lower than 60 m (200 ft) but not lower than 30 m (100 ft) and a runway visual range not less than 300 m;

iii) Category III (CAT III): a decision height lower than 30 m (100 ft) or no decision height and a runway visual range less than 300 m or no runway visual range limitations

4.3 Decision Altitude/Height: A specified altitude or height in a 3D instrument approach operation at which a missed approach must be initiated if the required visual reference to continue the approach has not been established.

Note: Decision altitude (DA) is referenced to mean sea level and decision height (DH) is referenced to the threshold elevation

4.4 ILS Critical Area: An area of defined dimensions about the localizer and glide path antennas where aircraft and vehicles are excluded during all ILS operations. The critical area is protected because the presence of vehicles / or aircraft inside its boundaries will cause unacceptable disturbance to the ILS signal-in-space.

4.5 ILS Sensitive Area: An area extending beyond the ILS critical area where the parking and/or movement of vehicles, including aircraft, are controlled to prevent the possibility of unacceptable interference to the ILS signal during ILS operations. The sensitive area is protected to provide protection against interferences cause by large moving objects outside the critical area but still normally within the airfield boundary

4.6 Low Visibility Procedures: Specific procedures applied at an aerodrome for the purpose of ensuring safe operations during Categories II/ III approaches and/or low visibility take-offs.

Note: As per para 5.3 of CAR on All Weather Operations, an operator shall not conduct Take-off with RVR/Visibility less than standard CAT-I conditions of 550m RVR/800m Visibility unless low visibility procedures are enforced.

4.7 Low Visibility Take-Off (LVTO): A term used in relation to flight operations referring to a take-off on a Runway where the RVR is less than 400m.

4.8 Manoeuvring Area: That part of an aerodrome to be used for the take-off, landing and taxiing of aircraft, excluding aprons.
4.9 Movement Area: That part of an aerodrome to be used for the take-off, landing and taxiing of aircraft, consisting of the manoeuvring area and the apron(s).
4.10Runway Visual Range: The range over which the pilot of an aircraft on the centreline of a runway can see the runway surface markings or the lights delineating the runway or identifying its centreline
4.11 Visibility (Doc 9365): Visibility for aeronautical purposes is the greater of:
a) the greatest distance at which a black object of suitable dimensions, situated near the ground, can be seen and recognized when observed against a bright background;
b) the greatest distance at which lights in the vicinity of 1 000 candelas can be seen and identified against an unlit background.
Note 1: The two distances have different values in air of a given extinction coefficient, and the latter b) varies with the background illumination. The former a) is represented by the meteorological optical range (MOR).
Note 2: The definition applies to the observations of visibility in local routine and special reports, to the observations of prevailing and minimum visibility reported in METAR and SPECI and to the observations of ground visibility
5 ABBREVIATIONS
5.1 The abbreviations used in descriptions of Low Visibility Procedures have the following meanings:
ARFF: AIRCRAFT RESCUE & FIRE FIGHTING
AGL: AERONAUTICAL GROUND LIGHT
AOM: AERODROME OPERATING MINIMA
ATC: AIR TRAFFIC CONTROL
ATIS: AUTOMATIC TERMINAL INFORMATION SERVICE
CFT: CRASH FIRE TENDER
DG: DIESEL GENERATING SET
GP: GLIDE PATH
ILS: INSTRUMENT LANDING SYSTEM
IHP: INTERMEDIATE HOLDING POSITION
LOC: LOCALIZER
LSA: LOCALIZER SENSITIVE AREA
LVP: LOW VISIBILITY PROCEDURE
MID: MID POINT
MET: METEOROLOGY
NAV: NAVIGATIONAL
RVR: RUNWAY VISUAL RANGE
SMC: SURFACE MOVEMENT CONTROL
SP: SAFEGUARDING PROCEDURES
SSO: SHIFT SUPERVISORY OFFICER
TDZ: TOUCHDOWN ZONE
TSO: TOWER SUPERVISORY OFFICER
WSO: WATCH SUPERVISORY OFFICER
6GENERAL
6.1The Low Visibility Procedure (LVP) incorporates safeguarding measures to mitigate runway incursions and defines operational restrictions to ensure safe Airside Operations taking into account the available Aerodrome facilities
6.2The provision mentioned in para 5.3 of DGCA Civil Aviation Requirements (CAR) (Rev.-10) section 8, Series ‘C’, Part-I on All-weather Operations necessitated the need of Low Visibility Procedures for accommodating/ permitting departures in Visibility/RVR less than 800M/550M at Imphal airport even though there are No CAT II and CAT III operations.
6.3Accordingly, Low visibility Procedures have been developed for Imphal Airport to accommodate/permit departures in Visibility/RVR less than 800M/550M up to 400 M RVR from RWY 04.
**7 MINIMUM REQUIREMENTS:**
7.1The following Aeronautical Ground lights and RVR equipment shall be serviceable to the required standard to support Low Visibility Procedures.
a) Runway edge lights,
b) Runway end lights,
c) Instrumented RVR at the beginning (TDZ RVR).
d) Stand by Power supply to maintain switch over time of 1 Second for Runway Edge Lights and Runway End Lights. This requirement can be met with the help of DG Set and/or UPS.
e) RWY Guard Lights
f) Road holding position light
g) Apron Flood Light
7.2Unserviceability of Aeronautical Ground Lights/ Equipment before Implementation of LVP
7.2.1The system of preventive maintenance employed for a precision approach runway category I shall have as its objective that, during any period of category I operations, all approach and runway lights are serviceable, and that in any event at least 85 per cent of the lights are serviceable in each of the following:
a) Precision approach category I lighting system;
b) Runway threshold lights;
c) Runway edge lights; and
d) Runway end lights.
In order to provide continuity of guidance an unserviceable light shall not be permitted adjacent to another unserviceable light unless the light spacing is significantly less than that specified
Note. — In barrettes and crossbars, guidance is not lost by having two adjacent unserviceable lights.
7.2.2The system of preventive maintenance employed for a runway meant for take-off in runway visual range conditions less than a value of 550 m shall have as its objective that, during any period of operations, all runway lights are serviceable and that in any event:
a) at least 95 per cent of the lights are serviceable in the runway centre line lights (where provided) and in the runway edge lights; and

b) at least 75 per cent of the lights are serviceable in the runway end lights.
In order to provide continuity of guidance, an unserviceable light shall not be permitted adjacent to another unserviceable light
7.2.3The system of preventive maintenance employed for a runway meant for take-off in runway visual range conditions of a value of 550 m or greater shall have as its objective that, during any period of operations, all runway lights are serviceable and that, in any event, at least 85 per cent of the lights are serviceable in the runway edge lights and runway end lights. In order to provide continuity of guidance, an unserviceable light shall not be permitted adjacent to another unserviceable light.
7.2.4During low visibility procedures the construction or maintenance activities in the proximity of aerodrome electrical systems shall be restricted.
Low Visibility Procedures will not be implemented when any of the light/equipment mentioned in para 7.1 above is un-serviceable or is not maintained as per the required standard.

| Aeronautical Ground Lighting Facility | Un-serviceability | Restrictions |
|---|---|---|
| Runway Edge lights | More than 5% of all lights are unserviceable | LVP operations will be suspended |
| | Any two consecutive lights or more are unserviceable | |
| Runway End lights | More than 25% of all lights are unserviceable | LVP operations will be suspended. |
| | Any two consecutive lights or more are unserviceable | |
| Standby Generators/UPS | Any of the generator/UPS is unserviceable | LVP operations will be suspended |
| RVR system | RVR system unserviceable | LVP operations will be suspended |

7.2.5Unserviceability of Aeronautical Ground Lights / Equipment after Implementation of LVP:
When any of the light/equipment mentioned in para 7.1 above becomes un-serviceable or fails to meet the required standard during periods of LVP, TWR shall advise the aircraft accordingly and LVP shall be suspended and information to this effect shall be included in ATIS broadcast.
8SAFEGUARDING PROCEDURES
8.1Safeguarding Procedures (SP) are instructions for relevant airport departments to prepare ground services and facilities for low visibility operations in order that when LVP are implemented all safeguarding procedures are complete. WSO/Duty Officer Tower will initiate and co-coordinate with all the concerned agencies for completion of safeguarding procedures before implementation of Low Visibility Procedures
8.2Safeguarding Procedures shall be initiated when:
a)The Visibility/RVR is less than 1200m and visibility/RVR is forecast to deteriorate to 800m or less; and/or
b)The cloud ceiling is less than 400ft and forecast to fall to 200ft or less.
8.3Whenever meteorological conditions stipulated in para 8.2 prevails or anticipated WSO shall coordinate with Airport Rescue Fire Station (ARFS) for implementing the Safeguarding Procedures (SP). WSO/Duty Officer Tower shall also coordinate with following agencies for initiation of implementation/ cancellation/completion of SP as appropriate.
i)Airport Operations
ii)Civil & Electrical Engg. /Powerhouse
iii)CNS Briefing/Equipment Room
iv)Duty MET Officer
v)Concerned Airlines
vi)CISF
8.4When all the concerned agencies have completed their necessary actions, they shall report to WSO/Duty Officer Tower that their Safeguarding Procedure (SP) is completed and the airport is safeguarded for LVP operations
8.5Actions to be taken by Various Agencies:
8.5.1Before commencement of winter season, a meeting will be held by Airport Director in the month of October every year to inform all airlines and agencies operating at airportabout their roles/ responsibilities and create awareness to ensure cooperation for safe airport operations during periods of low visibility.
8.5.2 All the agencies shall ensure that staff and drivers are suitably trained during Low visibility operations

8.6WSO/Duty Officer Tower shall ensure that the provision mentioned in para 8.7 are complied in respect of SP and proper log entry shall be made in respect of initiation of implementation/ cancellation/completion of SP as appropriate.
8.7 Safeguarding procedures include:
a)Positioning of CFTs at the predetermined positions (PDP).
b)Stopping of all maintenance works on the maneuvering area as well as removal of all men and mobile equipment from the said area.
c)Ensuring availability of secondary power supply for change over time of maximum one second for RWY Edge and RWY End lights supported by UPS
*RWY Edge and RWY End lights may continue to operate on main power supply during safeguarding Procedures. Whenever, LVP is to be implemented as per para 9 below, the RWY Edge and RWY End lights shall be put on Standby Power Supply (DG set or UPS). This operation needs to be completed before LVP is implemented.*
*As UPS is available at Imphal Airport and is capable of maintaining the required AGL system (refer table under para 7.2) with one second of Switch Over time with Main Supply, the main supply can continue to be primary supply and the Generator Supply can be kept as Standby Power supply. In case of UPS is unserviceable, Generator supply will become primary source of power supply and Main power supply shall act as standby power supply.*
d)Ensuring that Runways, Taxiways and other associated facilities such as Markings, Aeronautical Ground Lights are inspected within one hour preceding the implementation of LVP Operations. This is done to ensure that the lighting systems and markings are serviceable as described. Inspections should continue every 2 hours while LVP is in progress. These lighting inspections should be accorded priority and, if necessary, aircraft operations may be delayed.
e)Ensure all access roads leading to Runway are provided with signs, markings and are closed with chains.
f)Ensure Road-Holding Position Lights are serviceable

**9 LOW VISIBILITY PROCEDURES-TAKE OFF**

Implementation of Low Visibility Take off Procedures:
MET Officer shall inform WSO/Duty Officer Tower controller whenever Visibility/RVR reduces to 800 Meters or below and/ or cloud ceiling is at 200 ft or below. WSO/Duty officer Tower controller shall coordinate with all the agencies to confirm whether the Safeguarding procedures have been completed or not. When Visibility/RVR falls below 800M/550M and or Cloud Ceiling is 200 ft or below and safeguarding procedures are complete WSO/Duty Officer Tower will implement Low Visibility Procedures. WSO/Duty Officer Tower shall inform all users of the imposition of low visibility take off procedures and fill up the details of LVP implementation as given in Annexure - II
9.1A take-off alternate aerodrome shall be selected and specified in the operational flight plan if either the meteorological conditions at the aerodrome of departure are below the operator's established aerodrome landing minima for that operation or if it would not be possible to return to the aerodrome of departure for other reasons.
9.2Take-off minima established by the operator must be expressed as visibility or RVR limits, consideration all relevant factors for each aerodrome planned to be used and the aeroplane characteristics.
9.3 RVR/Visibility for Take-off (Commercial Transport Aeroplanes) is as per Table, below:

**Table: RVR/Visibility for Take-off (Commercial Transport Aeroplanes)**

| Take-off RVR/Visibility | |
|---|---|
| Facilities | RVR/VIS1 Cat A, B, C & D |
| Adequate Visual reference2 (Day only) | 500m |
| Runway edge lights or Runway Centre line markings3 | 400 m |
| Runway edge lights and Runway Centre line markings3 | 300 m |
| Runway edge lights and Runway Centre line lights | 200 m |
| Runway edge lights and Runway Centre line lights and relevant RVR information4 | 150m |
| High intensity Runway edge lights and Runway Centre line lights (spacing 15 m or less) and relevant RVR information4 | 125m |
| High intensity Runway edge lights and Runway Centre line lights (spacing 15 m or less), approved lateral guidance system and relevant RVR information4 | 75m |

Note 1 – The TDZ RVR/VIS may be assessed by the pilot
Note 2 - Adequate Visual reference means, that a pilot is able to continuously identify the take-off surface and maintain directional control.
Note 3 - For night operations at least runway edge lights or centre line lights and runway end lights are available.
Note 4 -The required RVR must be achieved for all relevant RVR reporting points (touchdown, mid- point and stop-end/roll-out). The governing RVR shall be the lowest of the reported RVRs
Note 5 - An operator shall not conduct low visibility take-offs in less than 400m RVR unless approved by DGCA

**10 ACTIONS BY VARIOUS UNITS DURING LVP**

10.1 Actions by ATC:
a)Inform CNS, Aircraft Rescue & Fire Fighting Services, Terminal Manager/Airside operations, Electrical, Civil, MT, CISF etc.
b)Only one aircraft shall be permitted to operate at a time on the maneuvering area during the time LVTO is in force.
c)Towing of aircraft during LVTO will be permitted only in case of extreme operational requirement. TWR shall ensure that the towing of aircraft is done under escort of "Follow Me" vehicles. "Follow Me" shall follow the route cleared by ATC.
d)TWR shall not permit any ground run on the maneuvering area except idle power run on the stands;
e)TWR shall ensure that "Follow Me" services are provided to pilots on request;
f)The number of the vehicles on the maneuvering area shall be restricted to bare minimum and records of all vehicles operating on the maneuvering area shall be maintained and all vehicles operating on the maneuvering area shall be in two-way contact with Aerodrome Control Tower.
g)The following may be included in ATIS. "LOW VISIBILITY PROCEDURES IN FORCE".
h)TWR shall permit departures only from the beginning of the Runway in use.
i)Whenever visibility/RVR is less than 800M/550M, Duty officer tower shall confirm from pilot that the reported RVR value is within minima before issuing take-off clearance
j)Duty Officer Tower shall stop all flight operations whenever RVR falls below 400 M.
k)Runway inspection should be done when interval between two aircraft movements is 30 minutes or more.
l)Inform changes in RVR readings to the departing aircraft as required.
m)Inform pilots of all failures of ILS, lighting system etc. relevant to ILS CAT I Low Visibility Operations.
n)Initiate emergency action if aircraft on CAT I ILS is not in radio contact, as expected.
o)Duty officer tower shall take the Take – off Alternate from pilot before issuing take-off clearance

**10.2Actions by CNS In-charge/ Duty Officer, NAV AIDS:**

a)On receipt of "Initiating SP" from ATC, CNS In-charge will inform the Duty Officer, NAV AIDS and have the ILS equipment and its status indicators in ATC units checked up. He will inform ATC of any unserviceability in the equipment which is likely to affect ILS CAT I operation.
b)On receipt of "Initiating LVP" from ATC that LVP are to be made effective CNS In-charge will maintain continuous watch on the performance of ILS equipment and will inform ATC of any un-serviceability which may affect ILS CAT I operation.

**10.3 Actions by Duty Manager, Airside operations/Terminal:**

Duty officer shall ensure that:
a)No vehicle/person enters or is present in the sensitive/critical areas of localizer and glide path.
b)All civil/electrical works in progress are to be stopped in the manoeuvring area immediately and men/material/ equipment to be removed from the sensitive/critical areas of localizer and glide path.
c)Duty Manager shall ensure closure of roads leading to critical/sensitive area by chains.
d)Ensure that Runways, Taxiways and other associated facilities such as Markings, Lights are inspected every 2 hours while LVP is in progress
NOTE:
i)After ensuring above, Follow-me vehicles/operational vehicles will confirm the same to the ATC.
ii)Subsequently, follow-me vehicle/ operational vehicles shall remain available and will maintain listening watch on Walkie-Talkie.
iii)Any information about unserviceability of any of the Runway visual aids or power supply system will immediately inform ATC Tower accordingly.
iv)No vehicles shall enter/cross in the vicinity of Runway without permission from ATC Control Tower. Such vehicles shall be able to communicate with ATC.
v)The vehicles of the ARFF services, civil, electrical division or of any other agency which are not equipped with RT but has to enter the Runway or taxiway for urgent operational requirement shall be escorted by the follow me jeep all the time.
e)On receipt of advice from ATC to implement Safeguarding procedure, the Terminal Duty Manager, will immediately inform to all airlines.

**10.4 Action by Shift Manager (Civil):**

a)Shift Manager (Civil) to ensure that all civil works in progress in manoeuvring area, are stopped and that the work area is restored in complete serviceable condition and confirm to WSO accordingly.

b)During CAT I operations, ATC may authorize operations of vehicles Specially trained about critical and sensitive area, on manoeuvring area, including the crossing of taxiways.

**10.5Action by Shift Manager (Electrical):**

a)On receipt of advice to implement Low Visibility Procedures from ATC, Shift Manager (Electrical) will check that following visual aids are serviceable and can be operated at full intensity:

i)Approach lighting system
ii)Approach supplementary lighting
iii)Runway edge lights
iv)Runway threshold and end lights
v)Runway Guard Lights
vi)Road-Holding Position Lights

NOTE: No adjustment in light intensity shall be made without permission from ATC Tower during LVP.

b)They shall ensure that no electrical maintenance works is carried out during LVP either in powerhouse or on any other electrical facilities used during CAT I operations.

c)They will inform the un-serviceability or any change in status of any facility/systems to ATC & Airside Ops Manager immediately.

d)AGL Inspections shall be carried out every 2 hours while LVP is in progress.

**10.6 Actions by ARFF:**

a)Positioning of crash fire tenders on pre-determined positions.

b)ARFF/ CFTs must obtain clearances from ATC prior to entry to any aircraft movement area other than the designated service roads.

**10.7 Actions by CISF:**

a)The officer in charge on duty shall ensure by deploying adequate manpower that vital electrical installations are properly secured and protected against any unauthorized intrusion.

b)The Officer in charge, on receipt of advice to implement Low Visibility Procedures, will immediately inform all access gates and CISF posts under their respective controls in operational area to restrict movement of vehicles under their control.

**10.8 Actions by Duty Officer (Meteorological Office):**

a)Duty Meteorological Officer would issue an 'Outlook for Low Visibility Procedures' to the ATC whenever he expects that the RVR and/or cloud ceiling will fall below 800 m and/or 400ft or less respectively.

b)Whenever Duty Meteorological Officer visualizes that RVR is likely to fall below 800 m and/or cloud ceiling to 200ft or less within next 2 hours, he will issue an 'Advisory Message' to ATC to this effect.

c)When the RVR and/or cloud ceiling are 800m and/or 200ft respectively and the trend is towards improvement in these elements of weather conditions the Duty Met Officer may, when requested by ATC, advise him about such improving weather conditions for the purpose of termination of LVP.

d)The Duty Met Officer shall be present in Control Tower during periods of reduced visibility and ensure that the RVR displays in the Control Tower is serviceable.

e)In case, Instrumented RVR is not available, real time RVR shall be provided, whenever requested.

**11TERMINATION OF LOW VISIBILITY TAKE OFF PROCEDURES**

11.1When Visibility/RVR improves to 800M/550 M or more and cloud ceiling is 200 feet or higher and trend is for improvement, WSO/Duty Officer Tower would terminate operations of LVTO. He may obtain advice from Duty Met. Officer regarding improvement in weather conditions before the termination of LVP. The WSO/Duty Officer Tower will intimate the following regarding termination of operations:

i)ARFF
ii)Terminal Manager/Operations, who will advise all the previously notified personnel to resume normal operations.
iii)CNS, Civil, Electrical In-charge
iv)M.T Pool
v)Duty Met Officer
vi)CISF

11.2 On cancelling of LVP, following message shall be included in two subsequent ATIS broadcasts. “LOW VISIBILITY PROCEDURES CANCELLED”.

11.3 If SP are implemented and LVP are not subsequently implemented and the visibility/RVR improves and is more than 1200 m and/or the cloud ceiling is 400ft or higher and both are forecast to remain above the required SP criteria, WSO/Duty Officer Tower may cancel SP.

**12 ACTIONS BY OTHER AGENCIES (AIRLINES, REFUELLING COMPANIES, CATERING AGENCIES, ETC.)**

12.1Every year before commencement of monsoon/winter season, a meeting will be held by Airport Director, to inform all airlines and agencies operating at airport about their roles/responsibilities and create awareness to ensure cooperation for safe airport operations during periods of low visibility.

12.2All the agencies shall ensure that staff and drivers are suitably trained during Low Visibility operations

12.3A refresher program for ATCO’s and personnel responsible for airside operations should be conducted every year.

12.4All agencies operating in the operational area shall ensure that only those vehicles that are absolutely essential for aircraft operations operate in the operational area during periods of low visibility. The drivers of these vehicles should keep a look out for taxiing aircraft and other vehicles to prevent accidents.

12.5All the vehicles must have their obstruction lights “ON” during Low Visibility Procedures operations.
12.6All instructions/sign boards provided for vehicular movement area/service roads must be followed while operating in the operational area.

## VEIM AD 2.21 NOISE ABATEMENT PROCEDURES

NIL

## VEIM AD 2.22 FLIGHT PROCEDURES

1.Circling approach not available at Imphal.

2.Standard Instrument Departures Procedures:

i.Standard Instrument Departures Procedures (SIDs) for Imphal have been developed for both Runways (RWY 04 and RWY 22).

ii.SIDs provide the track guidance upto interception of ATS route along with required obstacle clearance and thus eliminate the requirement of aircraft to climb over the VOR upto MSA before establishing on desired track.

iii.Due to the high terrain, the minimum climb gradient of 7% (425 FT/NM) has been specified as Procedure Design Gradient (PDG) until passing 8500 FT for all the SIDs.

iv.The minimum climb gradient of 3.3% (200 FT/NM) may be maintained after passing 8500 FT.

v.Speed restrictions have been specified with each SID wherever applicable.

vi.Altitude restrictions at the VOR/DME fix have been specified wherever necessary.

vii.All the radials and distances have been specified from Imphal VOR/DME (115.9 IIM).

viii.Any altitude restriction by ATC shall be imposed only after departing aircraft has reached the applicable minimum sector altitude (9500 FT or 11000 FT).

ix.Aircraft unable to follow the SIDs, shall inform ATC and request for alternate clearance.

## RADIO COMMUNICATION FAILURE PROCEDURES

### 1. INTRODUCTION

1.1 Radio communication failure procedures are described in Para 15.3 of ICAO PANS ATM DOC 4444. Based on these provisions, following radio communication failure procedures are established, to standardize the actions to be taken by the pilot of arriving and departing aircraft and ATCOs at Imphal International Airport.

### 2. GENERAL

2.1 All Transponder equipped aircraft experiencing Radio Communication Failure shall set Transponder to Mode A/C Code 7600 as soon as practicable to indicate that it has experienced air-ground communication failure. (Note: This Requirement of Setting transponder to Mode A/C code 7600 in no way imposes any restriction on the pilot's decision to set transponder to Mode A/C code 7500 or 7700, whenever required).

2.2 Immediately after detection of RCF, the aircraft shall attempt to establish communication with the appropriate air traffic control unit using all other available means (Imphal Tower – Main & Standby/Silchar Tower/Guwahati Control/Emergency frequency 121.5 MHz)

2.3 The RCF aircraft shall look for other traffic in aerodrome circuit and shall keep a watch for instructions as may be issued by visual signals by ATC.

2.4 Pilot shall make blind transmission to ATC of all necessary reports and actions taken by the aircraft, e.g. descent, turn, proceeding to waypoint, etc., irrespective of whether encountering partial/complete RCF.

2.5 As soon as it is known that two-way communication has failed, ATC shall take action to ascertain whether the aircraft is able to receive transmissions by various means such as using emergency frequency 121.5MHz, through other aircraft in contact with Imphal or through other adjacent ATS units and Guwahati ACC.

### 3. ASSIGNED RUNWAY AND ITS AVAILABILITY FOR RCF AIRCRAFT

3.1 In case of an arriving aircraft experiencing RCF, when Runway for landing has already been advised to the aircraft by ATC, such runway shall be considered as the assigned runway. If an arriving aircraft has not been advised of any runway, Runway 04 shall be considered as assigned runway for such arrival.

3.2 In case of departures from Imphal Airport is returning due to RCF, Runway 04 shall be considered as the assigned runway for landing, irrespective of the runway used for departure.

3.3 If Radio communication failure occurs after commencing instrument approach, the aircraft may continue its approach to land on such Runway.

3.4 Runway lights and Approach lights in 'SWITCHED ON' position shall indicate the availability of such Runway for aircraft experiencing RCF.

3.5 In case of strong tail wind conditions or any other reasons for assigned Runway, aircraft experiencing RCF will carry out a missed approach on assigned runway and after following complete missed approach for such approach, aircraft will climb to 11000 ft and commence Instrument approach procedure published for the runway which is in opposite direction to the assigned Runway. For example, Runway 22 in case of Runway 04 and vice versa.

### 4. PROCEDURES FOR ARRIVALS

4.1 In case Radio Communication failure takes place after establishing final approach track, aircraft may continue the approach and land if visual, or GO AROUND and carry out the missed approach procedure.

4.2 Except when descent clearance has already been received from Imphal ATC and acknowledged, pilot shall continue on ATS route cleared to IIM maintaining last assigned and acknowledged flight level by adjacent ATS unit. Commence descent to 11000 feet in holding procedure VOR IIM at or as close as possible to Expected approach time (EAT) last received and acknowledged; or if no Expected approach time (EAT) has been received and acknowledged, at or as close as possible to Estimated time of arrival (ETA) resulting from the current flight plan. Leave VOR IIM and carry out published IAP for assigned runway as per para 3.1.

4.3 Land, if possible, within 30 minutes after the estimated time of arrival (ETA) or the last acknowledged expected approach Time (EAT), whichever is later.

### 5. PROCEDURE FOR DEPARTURE

#### 5.1 DEPARTURE INTENDING TO CONTINUE TO DESTINATION

5.1.1 Any departing aircraft experiencing RCF and intending to continue to its filed Flight plan destination shall:

i) If following 'SID', climb to filed Flight Level with SID lateral and vertical restrictions to join ATS route and continue as per the filed flight plan.

ii) If following ATC DEP clearance (other than published SID), join ATS route maintaining cleared flight level/11000 ft whichever is higher. After joining ATS route, climb to filed Flight level and continue as per the filed flight plan.

#### 5.2 DEPARTURE INTENDING TO LAND BACK AT IMPHAL AIRPORT

5.2.1 Any departing aircraft experiencing RCF after departure and intending to land back at Imphal Airport, shall:

i) Three minutes after setting Mode A/C code 7600 or reaching 11000 ft or cleared Flight level (if higher than 11000 ft) whichever is later, take a turn to proceed direct to IIM VOR and Hold as published.

ii) Commence descend in the IIM Hold and then carryout published instrument approach procedure for Runway 04.

**VEIM AD 2.23 ADDITIONAL INFORMATION**

| 1. TWY details | |
|---|---|
| **TWY Designator** | **Remarks** |
| A | Link TWY between RWY and NCC hanger towards beginning of RWY22. |
| B | Link TWY between RWY and main Apron towards RWY 22 and ARP. |
| C | Link TWY between RWY and main Apron towards Fire station. |
| D | Link TWY between RWY and Isolation bay towards beginning of Fire station. |

2. Due to increased bird activity in and around airport, pilot to exercise caution while landing and takeoff.

3. **Exemption Granted by DGCA:**

DGCA India has granted exemption vide letter dated 01.09.2022 for non-compliant width of runway strip against provisions in CAR Section 4 Series B Part I Para 3.4.3 for the period up to 31.12.2024.

4. **Following intersection Take Off Run Available :**

| RWY | FROM TWY INT | TORA(M) | TODA(M) | ASDA(M) |
|---|---|---|---|---|
| RWY 04 | TWY B | 934 | 934 | 934 |
| RWY 04 | TWY C | 1083 | 1083 | 1083 |
| RWY 22 | TWY B | 1810 | 1810 | 1810 |
| RWY 22 | TWY C | 1661 | 1661 | 1661 |

## VEIM AD 2.24 CHARTS RELATED TO AN AERODROME

1. Aerodrome Chart (14 May 2026)
2. Aircraft Parking / Docking Chart (17 April 2025)
3. Aerodrome Obstacle Chart Type-A (Operating Limitations) RWY 04
4. Aerodrome Obstacle Chart Type-A (Operating Limitations) RWY 22
5. Instrument Approach Chart - ILS Procedure RWY 04
6. Instrument Approach Chart - VOR Procedure RWY 04 (27 November 2025)
7. Standard Departure Chart – Instrument (SID) RWY 04, IIM 4A (CAT A/B/C) (13 June 2024)
8. Standard Departure Chart – Instrument (SID) RWY 04, LIGSA 4A (CAT A/B/C) (13 June 2024)
9. Standard Departure Chart – Instrument (SID) RWY 04, MALNO 4A (CAT A/B/C) (13 June 2024)
10. Standard Departure Chart – Instrument (SID) RWY 22, KKU 4B, POREG 4B, LLP 4B (CAT A/B/C) (13 June 2024)
11. Standard Departure Chart – Instrument (SID) RWY 22, LIGSA 4K (CAT A/B/C) (13 June 2024)
12. Standard Departure Chart – Instrument (SID) RWY 22, MALNO 4B, LIGSA 4B (CAT A/B/C) (13 June 2024)
13. Standard Departure Chart – Instrument (SID) RWY 22, MALNO 4K (CAT A/B/C) (13 June 2024)
14. Standard Departure Chart – Instrument (SID), RNP1 (GNSS), RWY 04 CAT A/B/C (16 May 2024)
15. Standard Departure Chart – Instrument (SID), RNP1 (GNSS), RWY 04-Tabular Description (16 May 2024)
16. Standard Departure Chart – Instrument (SID), RNP1 (GNSS), RWY 22 CAT A/B/C (16 May 2024)
17. Standard Departure Chart – Instrument (SID), RNP1 (GNSS), RWY 22-Tabular Description 1 (16 May 2024)
18. Standard Departure Chart – Instrument (SID), RNP1 (GNSS), RWY 22-Tabular Description 2 (16 May 2024)
19. Instrument Approach Chart - VOR Y Procedure RWY 22 (17 April 2025)
20. Instrument Approach Chart - VOR Z Procedure RWY 22 (27 November 2025)

## VEIM AD 2.25 VISUAL SEGMENT SURFACE (VSS) PENETRATION

| Procedure | Procedure Minima | VSS Penetration |
|---|---|---|
| 1 | 2 | 3 |
| | | |